

Architectural Blueprint and Iterative Development of the DataSimple AI Teacher: Phase 2

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Foundational Architecture and Data Engineering: Phase 1 Retrospective

The transition from a traditional computer science mentality to a modern, data-driven workflow represents a significant psychological and technical hurdle for learners entering the analytics space. Historically, programming has been taught through a trial-by-fire methodology, focusing heavily on raw functions, loops, and abstract algorithms in isolation before ever introducing practical applications. The educational philosophy underpinning the DataSimple platform challenges this antiquated paradigm. The platform operates on the fundamental belief that learning data science with Python can be highly accessible, provided the student is guided through the difficult mental transition to a "coder mindset". By prioritizing practical, higher-level tools like Pandas, Scikit-Learn, and TensorFlow from the very beginning, the educational framework treats Python not merely as an abstract programming language, but as an interactive utility, akin to navigating an operating system with a computer mouse.

To encapsulate this practical, mentorship-oriented teaching style into a highly scalable digital format, an advanced Retrieval-Augmented Generation (RAG) chatbot was engineered during the initial development phase. This artificial intelligence teacher was explicitly designed to mimic the pedagogical approach of a supportive data science instructor, moving beyond answering generic coding questions by actively retrieving precise curriculum materials, guided projects, and methodologies taught directly within the proprietary ecosystem. The engineering of this platform required overcoming substantial hurdles in cloud architecture, data extraction pipelines, and advanced orchestration.

Cloud Economics and Vector Database Selection

Building an intelligent generative application represents merely the surface layer of modern machine learning engineering; constructing a scalable, highly automated, and economically viable data pipeline to feed that intelligence represents the true architectural challenge. For an educational platform, cost optimization dictates fundamental architectural choices. The explicit objective was to design a production-ready, highly scalable system that maintained a zero-cost standing footprint when not actively processing student queries.

When provisioning an AWS Bedrock Knowledge Base, the default configuration automatically deploys an Amazon OpenSearch Serverless collection as the underlying vector database. While OpenSearch is a highly robust, enterprise-grade search engine capable of indexing terabytes of unstructured corporate data, the analysis indicates it presents a massive financial trap for an educational platform.

OpenSearch Serverless bills for Minimum Compute Units simply to keep the database awake, indexed, and ready to answer potential queries, incurring a standing cost of roughly \$150 to \$250 per month even if exactly zero web traffic hits the application. Alternative native relational databases, such as Amazon Aurora PostgreSQL equipped with the pgvector extension, were also evaluated. However, establishing this infrastructure securely requires deploying a Virtual Private Cloud, configuring subnets, establishing strict security groups, and still incurs a steady monthly fee of \$40 to \$60 just to remain idle.

To circumvent these prohibitive cost structures, the architecture was fundamentally re-engineered to utilize Pinecone as the custom vector store backend.

Vector Database Solution	Infrastructure and Network Requirements	Monthly Standing Cost	Verdict for Educational RAG Workloads
Amazon OpenSearch Serverless	Automated via the AWS Bedrock Quick Create console.	~\$150-\$250 (Driven by Minimum Compute Units).	Extreme overkill; financially prohibitive for lightweight, low-traffic educational applications.
Amazon Aurora PostgreSQL (pgvector)	Requires extensive Virtual Private Cloud setup, subnets, security groups, and manual SQL initialization.	~\$40-\$60 (Idle instance compute time).	Demands complex network engineering and retains an unacceptable idle financial penalty.
Pinecone (Serverless Starter Tier)	API integration facilitated securely via AWS Secrets Manager. Native Bedrock support.	\$0.00 (Scales to true zero for up to 2GB of data).	The undisputed optimal choice. Lightning-fast, zero maintenance, and fully cost-optimized.

By securely locking the Pinecone API credentials within a digital vault using AWS Secrets Manager, Bedrock handles the computationally intensive tasks of chunking and embedding the educational Markdown files, while Pinecone acts as the silent, zero-cost storage layer. This highly optimized architecture costs exactly \$0.40 per month to securely hold the API key, ensuring the entire data engineering pipeline falls within the AWS Free Tier.

Data Extraction, Organization, and the Context Disconnect

The foundation of any high-performing Retrieval-Augmented Generation system is clean, semantically dense, and impeccably organized data. For the platform, this required extracting deep pedagogical knowledge from nearly 100 Guided Projects, Data Tips, and video lessons. However, the mission to extract, download, and organize these educational resources evolved into a significant engineering hurdle.

The initial data extraction strategy attempted to utilize a pre-release version of Python alongside the standard youtube-transcript-api library, which resulted in persistent, unresolvable crash loops. The architecture subsequently pivoted to yt-dlp, a highly robust downloading library, operating on a stable Python 3.12 build. While this successfully extracted official subtitle files, it introduced a secondary data engineering challenge regarding formatting. The raw files were explicitly designed for rendering text on video players and were heavily cluttered with useless metadata headers, hidden HTML tags, and continuous timestamp tokens. Injecting this unscrubbed formatting into a vector database would waste massive amounts of context window space, severely degrading retrieval performance. A custom Extract, Transform, Load pipeline script was developed to deploy Regular Expression loops, stripping away all timestamps and HTML tags to isolate the pure spoken words of the curriculum.

Organizing these files for optimal vector search required solving a structural issue known as the "Context Disconnect". Initially, the curriculum repository consisted of standalone transcript files and their corresponding Jupyter Notebooks. If uploaded separately, the semantic search algorithm might retrieve a conceptual explanation but completely fail to retrieve the runnable code block that demonstrated it, leading to incomplete answers or severe hallucinations. Furthermore, raw notebook files inherently possess excessive noise, bloated with execution counts and encoded image data. To rectify this, a Python script utilizing the nbconvert library was deployed. The script parsed the notebook structure, stripped out the execution metadata, transformed the remaining code into clean Markdown, and aggressively stitched it to the matching transcript to create a unified document. This merge ensured that the chunking algorithm would process the theoretical explanation immediately followed by the exact code that executed it, permanently and mathematically binding the context.

Serverless Frontend and the Limitations of a Monolithic Pipeline

To deliver this educational support, a secure serverless frontend architecture was established using an Amazon API Gateway as the public-facing interface and an

AWS Lambda function as the secure courier, interacting with a frontend constructed using Velo by Wix. The JavaScript was programmed to capture the user's text input, format it into a standardized JSON payload, and execute an asynchronous HTTP POST request directly to the API endpoint.

However, achieving true educational and business utility required dynamic state management to facilitate interactive practice tests and DataSimple project suggestion. By design, standard AWS Lambda functions are entirely stateless. If the artificial intelligence asked a multiple-choice question and the student replied with the single letter "a", the stateless function received the letter in a total vacuum, failing to retrieve relevant curriculum matches. To simulate conversational memory, the architecture implemented native session tracking, capturing the memory token from the AWS response and attaching it to all subsequent requests.

Even with session memory enabled, the system encountered a "Context Overwrite". When a student submitted an answer, the linear RetrieveAndGenerate API treated that incoming sentence as a new semantic search query. The model would read the newly fetched questions, become distracted, and completely forget it was supposed to be in a grading mode, effectively overwriting its own conversational memory. While frontend workarounds such as semantic camouflage, hidden context injection, and hyper-strict master prompting successfully stabilized the initial prototype, they ultimately exposed the hard limitations of a monolithic, single-agent architecture. Forcing one Lambda function to simultaneously act as a document retriever, a state-tracking test examiner, and a general tutor created a highly fragile system.

Phase 2 Setup: The Transition to Multi-Agent Orchestration

To achieve true enterprise scalability and permanently eradicate the Context Overwrite problem, the system architecture required a fundamental redesign. Phase 2 focused entirely on a pivot from a linear API to a Multi-Agent Orchestration model, where a supervisor agent would route request to either the Learn-Agent, to guide students to learning content, or the Test-Agent, designed to provide a short quiz. By decoupling these diverse responsibilities into specialized, isolated entities, the system gained the ability to natively manage conversational memory, selectively bypass database retrieval when it is not needed, and dynamically route tasks based on explicit intent.

Selecting the Orchestration Framework

When architecting the routing for the core platform features, the decision required evaluating dynamic conversational routing against visual pipeline building and strict deterministic state machines.

Orchestration Method	Core Mechanism and Benefit	Architectural Limitation or Tradeoff	Verdict for Educational Platform
Amazon Bedrock Agents	Natively supports a central Supervisor acting as a principal orchestrator, intelligently reading unstructured intent and routing to specialized sub-agents.	Requires high initial configuration complexity and strict role demarcation.	The clear winner for natural language routing and managing conversational memory.
Amazon Bedrock Flows	Visual builder linking models, prompts, and knowledge bases into sequential pipelines. Fantastic for rapid prototyping without heavy code.	Flows are generally acyclic and rely on predetermined branching paths. They struggle with dynamic, unpredictable conversational loops.	Insufficient for a fluid tutoring environment that requires continuous looping between grading and explaining.
AWS Step Functions	Highly deterministic, cyclic state machine. Provides unmatched reliability for infrastructure orchestration, error handling, and managing asynchronous wait times.	Too rigid for natural language conversational routing.	Exceptional for backend stability and ideal for future data processing pipelines, but not suited for frontline chat routing.

Amazon Bedrock Agents was selected to implement the Multi-Agent Collaboration pattern. This framework allows for the deployment of a Supervisor Agent to act as an intelligent dispatcher, seamlessly routing students to the retrieval-heavy Librarian for learning, and the retrieval-isolated Examiner for testing.

Defining the Specialized Sub-Agents

The core of Phase 2 involved creating two completely isolated agents in the console to handle specific functionalities. Anthropic Claude 3 Sonnet was selected as the foundation model across all agents to optimize for complex reasoning.

The Librarian (Learn) Agent was explicitly bound to the existing AWS Knowledge Base set up on Pinecone. The instructions were configured to strictly leverage the JSON metadata sidecar files to filter the database and deliver precise curriculum pathways. The persona was defined as Data Science Teacher Brandyn, with critical behavioral instructions to speak directly to the student as if on a one-on-one video call, maintaining a supportive and enthusiastic teaching tone. The agent was forbidden from sounding like a machine summarizing a textbook, and was commanded to use short paragraphs, simple bullet points, and ALL CAPS for emphasis rather than standard Markdown formatting.

Conversely, the Examiner (Test) Agent was created to operate completely independently of the Knowledge Base. The primary catalyst for this retrieval isolation was the strict 29-second AWS API Gateway timeout limit, which was being exceeded when the agent attempted to simultaneously grade and search the database for recommendations. By intentionally bypassing document retrieval, it entirely eliminates the "distracted retriever" hallucinations and token snowballing that plagued Phase 1. Instead, the system was engineered with a 21-question bank hardcoded directly into the prompt, featuring pre-written explanations that include specific DataSimple project and tip recommendations to guide students to the best place to practice. This entire question bank is positioned at the beginning of the prompt to natively take advantage of prompt caching features, which keeps the context warm and potentially saves up to 90% of the cost of input tokens. This isolation allows the Examiner to cleanly utilize native conversational memory to generate multiple-choice options, grade the student's answer, and loop through practice tests without losing context.

Friction Points: Technical Hurdles and UI Bugs

While the architectural blueprint for Phase 2 was sound, the actual setup and deployment encountered severe technical friction, spanning from deep orchestration trace bugs to notoriously stubborn AWS console user interface quirks.

The IAM Role Order of Operations Quirk

During the initial creation of the Librarian agent, I encountered a the order of

operations bug within the console interface. When setting up a new agent, the console provides an immediate option to attach a Knowledge Base. However, attempting to select the existing database and save the configuration resulted in an immediate backend error.

Investigation revealed that the error was not related to the database itself, but rather the timing of execution. Although the radio button for the execution role was selected, the platform had not actually created the Identity and Access Management role in the background. The system only mints the AmazonBedrockExecutionRoleForAgents security badge when the entire agent configuration is saved for the very first time. Because the role did not technically exist yet, the system threw an error when attempting to bind the database. The fix required ignoring the database section entirely, saving the blank agent to trigger the IAM creation, and then returning to the edit screen to attach the ground-truth data.

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Background Validations and Console Caching Loops

The setup of the Supervisor agent introduced a validation error regarding the Multi-agent collaboration toggle. When the collaboration toggle was activated, the console silently generated a blank collaborator row in the background, waiting for input. If the toggle was subsequently turned off without filling out the row, the user interface visually hid the section from the screen, but the backend validation system continued to read the hidden, empty row. Because the required fields were empty, it threw an error and permanently blocked the user from saving the agent. Resolving this required temporarily reactivating the toggle to force the UI to reveal the hidden fields, manually deleting the ghost data via the trash icon, and toggling it back off before saving.

Furthermore, the frontend frequently became trapped in permanent validation loops for fields that no longer existed. When the interface entered this stuck state, I determined that fighting the console was highly inefficient. I decided to abandon the broken edit screen entirely, deleting the stuck agent, and rebuilding the resource from scratch, a process that could be reliably executed in under 60 seconds.

Model Stealth Reversions and the Missing Brain Trap

The system exhibited a severe stability issue during version updates, where the console would silently revert the selected foundation model from Anthropic Claude 3.5 Sonnet back to the older Claude 3 Sonnet model. This reversion was obvious

because the older model struggled significantly with the complex XML formatting rules required for metadata extraction. Furthermore, this reversion frequently wiped the custom "Advanced Prompts" utilized for Knowledge Base response generation, resetting the agent to factory defaults and erasing the defining persona instructions.

This behavior was linked to an aggressive AWS optimization protocol dubbed the "Missing Brain" trap. The console displayed a critical warning: to optimize agent performance, the system does not use instructions if the agent has no action groups, relies on only one Knowledge Base, and has User Input disabled. Because the Librarian agent met all these conditions, the optimization protocols assumed it was a standard retrieval bot. To minimize latency, the system bypassed the orchestration loop entirely, turning the agent's internal reasoning brain off. It ignored the master prompt and simply piped the raw, unformatted vector search results directly back to the Supervisor. Re-enabling the "User Input" toggle successfully broke this short-circuit, forcing the agent's language model to remain active and process the strict formatting instructions on every conversational turn.

Timeouts, API Gateway Limits, and the Engine Swap

As the architecture expanded from a single agent to a multi-agent orchestration traces, the JSON payloads transmitted between the backend and the frontend became significantly more complex, introducing XML <answer_part> tags and sub-agent payloads. Testing the new architecture revealed the system was vulnerable to a timeout error.

When a simple greeting was sent, the system responded instantly. However, when a complex curriculum request was submitted, the frontend threw a 500 Internal Server Error. Diagnostic analysis proved that the AWS Lambda function was hitting its default 3-second timeout limit. The Supervisor required time to route the question to the Learn Agent, which then had to query the database, read the chunks, and format a response—a process taking approximately 5 to 8 seconds. Because Lambda terminated any script exceeding the 3-second threshold, the connection dropped before the payload could be returned. Modifying the Lambda configuration to extend the timeout limit to 60 seconds successfully stabilized the communication pipeline.

However, while extending the Lambda timeout to 60 seconds resolved the initial script termination, it exposed a subsequent, 29-second timeout limit on the AWS API Gateway. When the Test Agent attempted to grade the final question, search the database for recommendations, and format the output all in a single multi-agent relay step, it exceeded this 29-second threshold, causing the API Gateway to drop the connection and crash the pipeline. To resolve this critical bottleneck, the architecture required an "engine swap" specifically for the Test Agent. The

foundation model for the Examiner was transitioned from the heavy Claude 4.5 Sonnet model to the much faster Anthropic Claude 3 Haiku model to drastically reduce latency and prevent timeouts.

What Did Not Work: Orchestration Glitches and Hallucinations

The transition to a multi-agent framework introduced entirely new categories of logical failures. The interaction between autonomous agents required meticulous prompt engineering to prevent miscommunication and hallucination.

TOrchestration Agent: Prompt Summarization

The most critical orchestration failure observed during Phase 2 was the internal communication between agents. When the Supervisor evaluated a student's input to determine the routing path, it successfully identified that a curriculum request belonged to the Librarian agent. However, orchestration traces revealed a fundamental flaw in how the payload was transmitted.

If a user inputted, "can you recommend a good beginner data analysis tip", the Supervisor did not pass this raw string. Instead, it summarized the intent and translated it into a third-person system command, transmitting the payload: {"recipient":"Learn-v2","content":"The user is asking for a recommendation for a good beginner data analysis tip."}.

This unprompted translation completely broke the downstream agent's persona. The Librarian agent was strictly instructed to act like a teacher on a one-on-one video call with a student. However, because the incoming message was framed in third-person administrator terminology, the Librarian agent no longer believed it was interacting with a student. It deduced it was executing a backend administrative task for the Supervisor. Consequently, it dropped the warm pedagogical persona, ignored all formatting rules regarding capitalization and metadata extraction, and outputted a dry, generic summary of the database chunks.

This failure highlighted a fundamental principle of multi-agent systems: language models inherently desire to assist and summarize. When an agent is acting purely as a router, this helpfulness is fatal to the pipeline. To rectify this, the Supervisor required a strict "Raw Pass-Through" mandate, forbidding it from acting as a translator and demanding it serve as a literal wire passing exact, unedited student text.

Meta-Document Hallucination

A secondary retrieval hallucination emerged regarding the quality of the search results utilized by the Librarian agent. When the system was queried to find a "medium sklearn project", the Pinecone database performed its mathematical retrieval perfectly, finding the closest semantic vectors. However, rather than fetching the actual guided project notebooks containing the code, the trace logs revealed the system was retrieving "meta-documents".

These meta-documents were transcripts from introductory video lessons or syllabus overviews where the instructor discussed project lengths, genres, or potential ideas in the abstract. Because the vector database accurately matched the conversational filler words surrounding the query, it prioritized these abstract discussions over the dense technical code blocks. As a result, the agent hallucinated project suggestions based on passing conversational mentions rather than actual curriculum assets.

To fix this distraction, an "Agentic Search Query" instruction was implemented. Because Bedrock agents construct their own database search queries based on their system prompts, the Librarian was instructed to extract only the core technical keywords and explicitly ignore conversational filler, ensuring it prioritized retrieving code-heavy guided projects over introductory materials.

Test Agent Runaway Counting

The deployment of the Examiner agent revealed the inherent limitations of relying on generative models for structured counting. The Examiner was tasked with administering a short practice test. In an early iteration, after the student answered the final question, the agent was expected to grade it and conclude the test. However, diagnostic traces revealed a hilarious plot twist: the agent graded the final answer, got trapped in its own autoregressive generation loop, forgot it was supposed to stop, and autonomously generated an unauthorized fourth question.

Fascinatingly, the Supervisor agent intercepted this error. Because the Supervisor's prompt contained a passing mention that tests would consist of exactly three questions, it observed the Examiner's payload, realized the sub-agent had malfunctioned, and appended a message apologizing for the extra question before passing it to the user. While the Supervisor's error-correction was impressive, the runaway counting required a structural fix. The Examiner's instructions were heavily refactored to utilize silent XML scratchpads to track state meticulously, forcing the model to explicitly define its current mode before generating user-facing text.

What Worked: Prompt Evolution and Iterative Refinement

The successful stabilization of Phase 2 was driven entirely by meticulous, iterative prompt engineering. By analyzing the failures, the instructions for each agent were continuously refined, transforming basic conversational bots into highly structured, data-driven diagnostic tools.

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The Evolution of the Supervisor Agent

The Principal Routing Agent underwent five distinct architectural iterations, evolving from a simple delegation node to a complex logic controller.

V1: The Foundation of Routing: The initial configuration established the baseline requirement that the Supervisor must not answer questions personally, routing all queries directly to the solitary Librarian agent.

V2: Formalizing Transparency: To combat early context degradation, V2 introduced the critical routing rule, demanding the exact, unedited original input string be passed without summarization, establishing the Supervisor as a transparent pass-through layer.

V3: Expansion to Multi-Agent Management: V3 introduced the Examiner agent, establishing conditional routing logic. The Supervisor was instructed to utilize the Examiner only if the student explicitly requested a quiz or provided a single-letter answer corresponding to multiple-choice options.

V4: Structural Organization and Interception Protocols: V4 represented a massive leap in complexity, utilizing strict XML tags (<routing_table>, <critical_routing_rule>, <relay_protocol>) to enforce structure. This version orchestrated a pass between the Test and Learn agents. The logic dictated that if the Examiner agent returned a final grade containing a hidden <diagnostic_report> tag, the Supervisor must selectively break its transparency rule. It was required to extract the report, delete the XML to hide it from the user, send the diagnostic text to the Librarian agent as an autonomous search query, and append the resulting project recommendations to the bottom of the final grade, delivering a single cohesive message.

V5: Functional Specificity and Optimization: While the Relay Protocol in V4 was an architectural masterpiece, execution traces revealed that pinging a second agent and waiting for a second database search incurred severe latency penalties. V5 optimized the system by stripping out the interception protocol, maintaining the XML structure but redefining the specific capabilities of each agent. By eliminating

the multi-step pass through, the V5 architecture ensured rapid response times.

Frontend Abstraction: A major architectural benefit of this finalized Supervisor agent is that it acts as a perfect abstraction layer for the Wix Velo frontend. By connecting the Wix website directly to the Supervisor, the frontend code never has to know about or manage the underlying Learn and Test sub-agents, completely shielding the user interface from backend complexity.

The Evolution of the Examiner (Test) Agent

The Examiner agent was systematically refined to ensure high-quality assessments and eliminate the runaway counting glitch.

V1: Basic Memory and Grading: Relying on general conversational memory, the agent searched the knowledge base, formulated questions, and graded responses by providing a 2-3 sentence explanation of why an answer was correct.

V2: The XML Scratchpad: V2 introduced significant internal reasoning upgrades to prevent state-loss. The concept of "The Scratchpad" forced the agent to use silent XML <thinking> tags to count and plan its state before responding. It also introduced the generation of a hidden <diagnostic_report> tag to summarize the student's performance for the routing agent.

V3: Strict Subject Distribution and Localized Recommendations: To ensure a comprehensive evaluation, V3 replaced general knowledge searches with a strict curriculum order: Question 1 must cover Data Analysis, Question 2 Machine Learning, and Question 3 Deep Learning. Furthermore, to solve the latency issues caused by the Supervisor's Relay Protocol, V3 offloaded the recommendation engine entirely onto the Test Agent. The Test agent was instructed to search its own connected knowledge base to find exactly two relevant projects based on the student's incorrect answers, enforcing the -- -- metadata extraction formatting.

V4: The Static Question Bank: To exert absolute quality control over the assessments and eliminate generative hallucinations, V4 shifted the agent from generating questions dynamically via vector search to selecting them from a hardcoded, highly verified <question_bank> provided directly in the prompt. The recommendation logic was refined to summarize the specific projects that were already embedded within the pre-written explanations of the test questions, maximizing relevance and minimizing execution time. Furthermore, hardcoding this 21 question bank directly into the Test Agent's system prompt (rather than querying the vector database) allowed the system to natively trigger AWS Bedrock's automatic prompt caching. Because the prompt exceeded the 1,024-token minimum requirement, AWS automatically cached the question bank and kept it "warm" over a 5-minute sliding window of inactivity, which maximized execution

speed and reducing processing costs.

The Librarian (Learn) Agent: Guardrails and Metadata

The prompt engineering for the Librarian agent focused on enforcing strict pedagogical guardrails and standardizing the output formatting to ensure a pristine user experience.

The "Two Projects" rule was hardcoded to prevent choice paralysis, mathematically constraining the agent to provide exactly two distinct suggestions whenever a student asked for recommendations, alongside a brief explanation of why the project fit their learning journey.

Crucially, the Metadata Extraction rule was implemented to eliminate hallucinations. The agent was forbidden from inventing project titles or recommending external platforms. To find the official name of a lesson, it was mandated to look at the hidden metadataAttributes block attached to the search results, extract the exact string from the title field, and output it using -- -- formatting. If the knowledge base lacked a specific answer, the agent triggered an "Out of Bounds" response, redirecting focus to the established curriculum.

The agent was also tasked with actively cleaning the raw data outputs, stripping out residual YouTube video IDs that survived the Phase 1 extraction pipeline, and ensuring Python code was only displayed when it served as a direct answer to a complex inquiry.

Conclusion: Key Learnings and the Path Forward

The architectural development of Phase 2 yielded critical insights into the orchestration of large language models within serverless environments. The most profound realization was the absolute necessity of retrieval isolation. The monolithic pipeline of Phase 1 failed because it forced state-tracking and vector retrieval to happen simultaneously, resulting in context overwrites. Decoupling the system into specialized nodes—specifically, untethering the Examiner agent from continuous document fetching—proved that native conversational memory is drastically more stable in isolation.

Furthermore, the debugging of the orchestration traces highlighted a fundamental friction point in multi-agent networks: language models possess an inherent bias toward summarizing and assisting. When an agent is deployed purely as a router, this helpfulness is highly destructive. The strict enforcement of a "transparent pass-through layer," explicitly forbidding intent translation, is a mandatory architectural pattern to prevent intermediary nodes from diluting user intent and

breaking the downstream execution of specialized personas.

Navigating the AWS Bedrock console also required adaptation. I struggled with finding a way around undocumented UI validation loops. The adoption of a "Clean Build Strategy"—deleting stuck resources and recreating them from scratch in under 60 seconds—proved to be the most resilient approach to bypassing phantom caching bugs and IAM role generation delays.

The successful stabilization of Phase 2 establishes a highly robust foundation for the planned expansions of the educational platform. With the Supervisor, Librarian, and Examiner agents functioning seamlessly in a synchronized orchestration loop, Phase 3 will focus on introducing the Debugger Agent. By equipping this new entity with API action groups to search the live internet, it will troubleshoot broken student code against current library documentation in real-time, completely decoupled from the historical Pinecone database. Finally, Phase 4 will implement AWS Step Functions and DynamoDB to establish a serverless, highly deterministic interactive student feedback loop, ensuring the continuous, data-driven evolution of the curriculum and the platform's overarching architecture.